MILITARY SPECIFICATION

FIN ASSEMBLY, M158

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies for the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements and provides quality assurance provisions for the Fin Assembly, M158 for use on 81MM Mortar Ammunition.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-H-6088	- Heat Treatment of Aluminum Alloys
MIL-A-48078	- Ammunition Standard Quality Assurance Provisions, General Specification for

FSC 1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1169 - Packaging, Packing and Marking for Shipment of Inert Ammunition Components

2.1.2 Other government documents, drawings and publications. The following other documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9221071 - Fin Assembly, M158

PUBLICATIONS

ANSI B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standard Institute (ANSI), 1430 Broadway, New York, N.Y. 10018.)

ASTM Method E-8 - Tension Testing of Metallic Materials
ASTM Method E-340 - Macroetching Metals and Alloys

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19105).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies).

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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2.1.3 Order of precedence. In the event of a conflict between the text of this specification and references cited herein, (except drawings) the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications. Materials shall not exhibit defects such as cracks, splits, porosity, etc.

3.2 Components and assemblies. - The components and assemblies shall comply with all requirements specified on drawing (dwg.) 9221071 and associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Heat treatment. - Heat treatment equipment, procedures and controls shall be in accordance with MIL-H-6088. The material shall be solution heat treated in such a manner as to avoid overheating of the material or cracking during quenching.

3.4 Hydrostatic test. - The fin housing shall be able to withstand a hydrostatic load of 17,250 ± 250 p.s.i. when tested in accordance with 4.5.4. Each fin housing that successfully meets the requirements of this test shall be permanently stamped with the letter "H" .125 inch maximum height by .01 inch maximum depth on the outside surface following the nomenclature.

3.5 Data requirements. - The contractor shall generate data in accordance with the requirements of the data item descriptions cited herein and in 4.5.3.3, and 4.5.4.

3.6 First article inspection. - This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Workmanship

3.7.1 Metal characteristics. - All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, pipe type defects, planar type defects, inclusions, porosity, or any similar characteristics.

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3.7.2 Threads. - Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings.

3.7.3 Burrs. - No part shall have a burr which might interfere with the assembly or function of the round or which might be injurious to personnel handling the item.

3.7.4 Foreign matter. - All parts and assemblies shall be free of chips, dirt, grease and other foreign matter that may impair proper functioning. The cleaning method used shall not be injurious to any of the parts, nor shall the parts be contaminated.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspections shall be conducted on this item:

a. First article inspection

b. Quality conformance inspection

4.3 First article inspection

4.3.1 Submission. - The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated (See Table 1).

Part description	Drawings	Quantity
Fin Assembly, M158	9221071	10
Fin	8882167	5
Housing	9212304	5
Housing, prior to drilling of flash holes		10
Increment holder (rear)	9221072	5
Increment holder (front)	9220708	5

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Following nominal cropping, one specimen suitable for macroetch testing shall be submitted from the rear of both housing and fin stock. In addition, 4 tensile specimens shall be submitted from fully heat treated material. Two tensile specimens shall be obtained from rolled, extruded, or forged housing material stock (as applicable), and two specimens shall be obtained from fin stock. See Table 1, Note a. Microstructure specimens shall be taken from both fin and housing material stock as described in 4.5.3.2.

4.3.2 Inspections to be Performed - See MIL-A-48078 and Table 1 specified herein.

4.3.3 Rejection - See MIL-A-48078.

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TABLE I. First article inspection

MIL-F-48076A(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 3		DRAWING NUMBER
	Fin, M158, Components and assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fin assembly (dwg. 9221071) Examination for defects	10	100%	3.2	4.4.2.7
	Fin (dwg. 8882167) Examination for defects	5	100%	3.2	4.4.2.6
	Housing, Cartridge (dwg. 9212304) Examination for defects	5	100%	3.2	4.4.2.5
	Housing, Cartridge, prior to drilling flasholes Hydrotest	10	100%	3.4	4.5.4
	Increment holder (rear) dwg. 9221072 Examination for defects	5	100%	3.2	4.4.2.3
	Increment holder (front) dwg. 9220708 Examination for defects	5	100%	3.2	4.4.2.3

NOTES:

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 3 OF 3		DRAWING NUMBER
	Fin, M158, Components and assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fin stock (press-quenched type only) after extrusion				
	Macroetch test	2	100%	3.1	4.5.2/visual
	Microstructure specimen analysis	2a/	100%	3.3	4.5.3.2/visual
	Tensile test	2	100%	3.1	4.5.3.1.2/visual

NOTES:
a/ Microstructure specimens shall be taken from an underformed portion of each tensile specimen after the tension test is performed.

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4.4 Quality Conformance Inspection

4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078 and the following as applicable:

4.4.1.1 Cartridge housing lots. - A cartridge housing lot shall consist of material of the same alloy that has been heat treated in the same heat treating equipment at the same temperature in one unchanged process. The maximum number of pounds in a cartridge housing lot shall consist of 15,000 or fraction thereof. A cartridge housing lot may consist of more than one solution heat treatment batch. A solution heat treat batch shall consist of not more than 2000 pounds of the same alloy that has been thermally treated at the same time by the same method in the same furnace equipment at the same time and temperature in one unchanged process. Tensile and microstructure tests hereinafter prescribed must be selected from and represent each solution heat treatment batch. The identification and segregation of each solution heat treatment batch is required until compliance with the tensile and microstructure requirements is confirmed; at that time the above mentioned segregation is no longer necessary.

4.4.1.2 Fin stock lots. - The fin stock lot shall consist of not more than 2000 pounds of extruded stock of the same alloy that has been thermally treated by the same method in the same furnace equipment at the same temperature and time in one unchanged process regardless of the solution heat treating method utilized. Each fin stock lot representing material that has been solution heat treated after extrusion, must be solution heat treated at the same time in the same furnace using the same temperature and time in that furnace. The fin stock lot thus constituted shall be the basis for selecting the sample blanks for the tension and microstructure tests hereinafter prescribed. Fin stock lots produced by the press-quench method may be up to 2,000 pounds each or fraction thereof.

4.4.2. Examination - See MIL-A-48078.

a. Sampling plans. - Sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Tool control dimensions. - Dimensions marked tool control shall be inspected as follows: select and gage

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one fin and one housing from each machine in operation every two hours or fraction thereof and whenever tooling is changed. Prior to any tooling change, the last housing or fin produced from the old tooling shall be gaged for the requirements of the applicable drawing. All material produced prior to periodic inspection shall be rejected and screened whenever a defective tool control dimension is found and corrective action must be taken.

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QUALITY CONFORMANCE INSPECTION

MIL-F-48076A(AR)

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Ignition cartridge housing, bar stock for				NEXT HIGHER ASSEMBLY 9212304
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
A	Macroetch examination	2	100%	3.1	4.5.2/visual
B	Tension test	2	100%	3.1	4.5.3.1.1/test
C	Microstructure examination	2a/	100%	3.3	4.5.3.2/visual
Major	None defined				
Minor	None defined				

NOTES: a/ Microstructure specimens shall be taken from an undeformed portion of each tensile specimen after the tension test is performed, e.g. cross-section of the threaded portion.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48076A(AR)

PARAGRAPH	TITLE		SHEET 1 OF		DRAWING NUMBER
4.4.2.2	Fins, bar stock for				NEXT HIGHER ASSEMBLY 8882167
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
A	Macroetch examination	2	100%	3.1	4.5.2/visual
B	Tension test	2	100%	3.1	4.5.3.1.2/test
C	Microstructure examination	2a/	100%	3.3	4.5.3.2/visual

NOTES: a/ Microstructure specimens shall be taken from an undformed portion of each tensile specimen after the tension test is performed.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9220708/9221072
4.4.2.3	Holder, Increment, Front and Rear				NEXT HIGHER ASSEMBLY 9221071
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of hole		0.40%	3.2	Gage
102	Thickness		0.40%	3.2	Gage
103	Location of fingers		0.40%	3.2	Gage
104	Diameter of hub		0.40%	3.2	Gage
Minor					
201	Length of fingers		0.65	3.2	Gage
202	Width of fingers		0.65	3.2	Gage
203	Radii missing or improper		0.65	3.2	Gage
204	Evidence of poor workmanship		0.65	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Housing, Cartridge, prior to drilling flash-holes (19)		9212304
			NEXT HIGHER ASSEMBLY 9221071

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Special					
A	Hydrotest		100%	3.4	4.5.4/See 6.9
Major	None defined				
Minor	None defined				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9212304
4.4.2.5	Housing, Cartridge				NEXT HIGHER ASSEMBLY 9221071
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	External thread length, min.		0.40%	3.2	Gage
102	External thread pitch diameter (PD)		0.40%	3.2	Gage
103	External thread major diameter, min		100%	3.2	Gage
104	Internal thread PD		0.40%	3.2	Gage
105	Internal thread length, min		0.40%	3.2	Gage
106	Diameter of ignition cartridge cavity		0.40%	3.2	Gage
107	Diameter of radial flashholes		0.40%	3.2	Gage
108	Diameter of primer flashhole		0.40%	3.2	Gage
109	Location of flashholes		0.40%	3.2	Gage
110	Outside diameter forward of taper		0.40%	3.2	Gage
111	Total length		0.40%	3.2	Gage
112	Internal thread minor diameter, max.		100%	3.2	Gage
113	True position of outside diameter with PD of forward threads		0.40%	3.2	Gage
114	True position of ignition cartridge cavity with PD of forward threads		0.40%	3.2	Gage
115	Depth of ignition cartridge cavity		0.40%	3.2	Gage
116	True position of primer flashhole with ignition cartridge cavity		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9212304
4.4.2.5	Housing, Cartridge				NEXT HIGHER ASSEMBLY 9221071
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Outside diameter at shoulder		0.65%	3.2	Gage
202	Distance from rear face to shoulder		0.65%	3.2	Gage
203	Radius, chamfer or taper missing or inadequate		0.65%	3.2	Visual/6.6
204	Countersink missing from rear face		0.65%	3.2	Visual
205	True position of primer flashhole with ignition cartridge cavity		0.65%	3.2	Gage
206	Evidence of poor workmanship		0.65%	3.7	Visual
NOTES:					

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.6	Fin				8882167
					NEXT HIGHER ASSEMBLY 9221071
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Forward outside diameter of blades		0.40%	3.2	Gage
102	Rear outside diameter of blades		0.40%	3.2	Gage
103	Pitch diameter of rear threads		0.40%	3.2	Gage
104	Minor diameter of rear threads		0.40%	3.2	Gage
105	Pitch diameter of forward threads		0.40%	3.2	Gage
106	Minor diameter of forward threads, max.		100%	3.2	Gage
107	Angle between fin blades		0.40%	3.2	Gage
108	Total length		0.40%	3.2	Gage
109	Thickness of fin blades (6)		0.40%	3.2	Gage
110	Length of rear thread, min		0.40%	3.2	Gage
111	Length of forward thread, min		0.40%	3.2	Gage
112	Length from rear face to forward face		0.40%	3.2	Gage
113	Hub diameter		0.40%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 8882167
4.4.2.6	Fin, continued				NEXT HIGHER ASSEMBLY 9221071
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
114	Diameter of forward counter bore		0.40%	3.2	Gage
115	Depth of forward counter bore		0.40%	3.2	Gage
116	Angle at front of fin blades		0.40%	3.2	Gage
117	Height of blade cant (6)		0.40%	3.2	Gage
118	Length from rear of blades to end of blade cant		0.40%	3.2	Gage
119	Blades canted in wrong direction		0.40%	3.2	Visual
Minor					
201	Length of rear outside diameter		0.65%	3.2	Gage
202	Angle at rear of forward outside diameter		0.65%	3.2	Gage
203	Outside diameter at front of forward taper		0.65%	3.2	Gage
204	Diameter of rear counterbore		0.65%	3.2	Gage
205	Surface finish improper		0.65%	3.2	Visual/6.7
206	Radii or chamfers missing or incorrect		0.65%	3.2	Visual
207	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.7	Fin Assembly, 81	Cartridge: M158			9221071
					NEXT HIGHER ASSEMBLY
					9220705
CATEGORY	EXAI	ATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical						
1	Primer flashhole	missing or obstructed		100%	3.2	Gage
Major						
101	Radial flashhole	s) missing or obstructed (18)		100%	3.2	Gage/4.5.5.1
102	Rear increment h	der out of position		0.40%	3.2	Visual
103	Fin blade(s) not	parallel with forward thread		0.40%	3.2	Gage
104	Fin blade(s) not	straight up to cant		0.40%	3.2	Gage
105	Fin blade(s) not	perpendicular to front face		0.40%	3.2	Gage
106	Algebraic total	f all blades exceeded		0.40%	3.2	Gage
107	Flashholes exces	ive in number		0.40%	3.2	Visual/4.5.5.2
108	True position of	rear thread with forward thread a	d front face	0.40%	3.2	Gage
109	True position of	outside diameter of fin blades with	orward thread and front face	0.40%	3.2	Gage
110	Fin blade(s) bro	en or cracked		0.40%	3.2	Visual
111	Torque inadequat		3a/		3.2	Test/4.5.6
112	Sealing compound	inadequate in quality or quant	ty 3b/		3.2	Test/4.5.7
113	Rear increment h	lder loose		0.40%	3.2	Manual

NOTES:
a/ Use the same sampl s for torque test and sealing compound adequacy test.
b/ Test should be perform d only after compound has fully cured.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48076A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.7	Fin Assembly, 81MM Cartridge: M158, cont.				9221071
					NEXT HIGHER ASSEMBLY 9220705
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Sealing compound on forward or rear threads		0.65%	3.2	Visual
202	Lot number, date, nomenclature, or "H" missing or improperly stamped		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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4.4.3 Testing. - Testing is described in the First Article and Quality Conformance Inspection tables.

4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection tables herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

4.5.1 Materials, components and processes. - Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and mechanical inspections. - The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein and Table I.

4.5.2 Macroetch examination. - A full section macroetch specimen shall be cut from each end of the bar representing the first and last extruded material from each individual extrusion billet. Failure to comply with the requirements of 3.7.1 will require an additional length of discard from the defective extrusion end(s) and testing of an additional macroetch specimen(s) taken from the end of the remaining length of extruded material. This procedure must be continued until a sufficient length of defective extrusion has been discarded and the requirements of 3.7.1 are met. Suitable identification of each length of extrusion and corresponding macroetch must be maintained until the extruded material is accepted. The specimen shall be macroetched using the criteria prescribed in ASTM Method E-340 except that sodium hydroxide shall be used as the macroetching reagent.

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4.5.3 Mechanical properties

4.5.3.1 Tension test

4.5.3.1.1 Cartridge housing lot. - Two specimens, one from the hottest and one from the coldest zone of each solution heat treatment batch shall be subjected to this test. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Specimens taken shall conform to the largest obtainable round type specimen as prescribed by ASTM Method E-8 and approved by the procuring agency; the Government shall not be restricted from testing additional specimens to determine that the mechanical properties conform to drawing requirements. The criteria for performing this test shall be as prescribed by ASTM Method E-8.

4.5.3.1.2 Fin stock extrusion lots. From press-quenched fin stock (only), two specimens total shall be selected as follows: one specimen shall be taken from both the front and rear of each extrusion lot (except areas cropped in accordance with macroetch examination) representing the first and last material respectively which has been extruded and is in accordance with the provisions of 4.4.1.2. Specimens from fin lots made by solution heat treatment following extrusion (only) shall be taken as follows: two specimens total shall be selected; one shall be taken from the hottest zone and the other from the coldest zone of each solution heat treatment batch which is a part of a lot in accordance with the provisions of 4.4.1.2. Failure of any tension test specimen to comply with the yield strength or elongation requirements of the applicable drawing shall be cause for rejection of the lot. Specimens taken shall conform to the largest obtainable flat type specimen as prescribed by ASTM Method E-8 and approved by the procuring agency; the Government shall not be restricted from testing additional specimens to determine that the mechanical properties conform to drawing requirements. The criteria for performing this test shall be prescribed by ASTM Method E-8.

4.5.3.2 Microstructure examination. - From each of the specimens submitted for tension testing in 4.5.3.1. and 4.3.1, one wafer, approximately one-half inch in length shall be cut longitudinally and prepared for metallographic

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examination. When so prepared, the specimen shall be etched with Keller's reagent and examined for metallurgical defects by scanning under a suitable microscope at a minimum magnification of 500 diameters. Overheating shall be evidenced by the presence of rosettes in the microstructure or incipient melting at the grain boundaries (see 6.5). Evidence of the above shall cause rejection of the representative solution heat treatment batch.

4.5.3.3 Quality inspection report. - The contractor shall submit a quality inspection test, demonstration, and evaluation report in accordance with Data Item Description DI-R-1724 (tailored) giving the detailed test results from the above tests (4.5.3.1 and 4.5.3.2).

4.5.4 Hydrostatic test. - After final machining but prior to drilling flash holes and prior to gaging, each cartridge housing shall be subjected to a hydrostatic test to determine compliance with the requirements of 3.4 and 3.7.1. Failure to meet the requirements of 3.4 and 3.7.1 shall be cause for rejection of the housing. The housing shall be tested using equipment capable of performing the test accurately and consistently. In performing the test, the pressure will be applied gradually until the required minimum pressure is reached and held for three seconds. The equipment shall be calibrated prior to the start of each day's operation and at intervals of not over 4 hours of continuous operation. It shall also be calibrated prior to restarting the operation after a lay-off of three hours or more. If it has been found that the equipment is out of calibration in such a direction that the pressure was actually lower than required, the housings tested since the last satisfactory calibration shall be retested after the calibration error has been corrected. Rupture, cracking, or bulging shall be evidence of failure and cause for rejection of the housing. Each housing which has successfully passed the test shall be stamped in accordance with 3.4. The contractor shall submit a monthly quality defect report in accordance with Data Item Description DI-R-1721 summarizing the defects found.

4.5.5 Flash hole inspection

4.5.5.1 Missing flash holes. - Each fin housing shall be inspected by a non-visual system capable of determining the presence of the required number of flash holes (see drawing).

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4.5.5.2 Excess flash holes. - Each fin housing shall be inspected by a method capable of determining the presence of flash holes in excess of the drawing requirement.

4.5.6 Torque test. - Random samples supplied in accordance with 4.4.2.7 shall be tested by applying the required assembly torque to the fin in the direction of assembly before the thread sealing compound has set. Any perceptible movement of any fin will be considered as failure of the assembly to comply with the drawing requirement and shall be cause for 100% back inspection to the last successful test. The joint shall be marked to provide a means for detection of movement of the fin with respect to the cartridge housing. After testing, the parts shall be set aside for disassembly testing after the thread sealing compound has cured. The torque test shall be performed daily.

4.5.7 Sealing compound test. - This test shall be performed daily after the random samples used for the torque test are set aside to allow the thread sealing compound to cure. An additional set of 3 samples shall be taken if a new batch of sealing compound is used after the day's production has started. The samples supplied in accordance with 4.4.2.7 shall be tested by applying 225 $\pm$ 25 in-lbs disassembly torque to the fins after the thread sealing compound has cured. Any perceptible movement of any fin shall be cause for rejection of the sealing compound mix. If the movement occured as the result of inadequate application of sealing compound, 100% back inspection to the last successful test shall be performed.

5. PACKAGING

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with the requirements of level C in MIL-STD-1169.

6. NOTES

6.1 Intended use. - The components covered by this specification are intended for use on the mortar, 81MM, Illuminating, M301A3. This is a pressure vessel application. The contractor may find it cost effective to perform additional non-destructive inspections (such as ultrasonic inspection) of the bar stock prior to machining and hydrostatic testing. Costs incurred as the result of

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additional tests shall not be reimbursed by the Government. In no case shall the performance of additional tests relieve the contractor from complying with any of the regular test requirements contained in this specification.

6.2 Ordering data. - See MIL-A-48078.

6.3 Submission of inspection equipment for design approvals. - See MIL-A-48078. Submit designs as required to Commander, U.S. Army Armament Research and Development Center, ATTN: DRSMC-QAT-I(D), Dover, NJ 07801

6.4 Drawings. - Drawings listed in Section 2 of this specification under the headings U.S. Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal and U.S. Armament Research and Development Command (ARRADCOM) drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Microstructure examination. - Microstructure examination of solution heat treat specimens should be performed by a skilled metallographer familiar with aluminum alloys.

6.6 Visual examination qualification. - When compliance with the applicable requirements is in doubt as a result of visual examination, these characteristics may be measured or gaged to determine acceptability.

6.7 Surface roughness. - The roughness comparison specimens prescribed by ANSI-B46.1 shall be used as a basis for surface roughness determination. Four power (4X) magnification may be used in performing visual examination.

6.8 Data requirements. - Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirements	Applicable DID
4.5.3.3	Quality Inspection Test, Demonstration and Evaluation Report	DI-R-1724 (tailored)
4.5.4	Quality Inspection Defect Report	DI-R-1721

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(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.9 Hydrotest. Use of the threads for a hydrostatic seal is not mandatory and may result in an adverse change of the thread form. The hydrotest may be performed on the housing prior to machining the threads if the wall thickness is in conformance with the drawing requirement.

6.10 Changes From Previous Issue. - Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-48076A (AR)	Fin Assembly, M158

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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